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An Information Theoretic Approach for Identifying Shared Information and Asymmetric Relationships Among Variables

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Behavioral researchers are often faced with the need to identify complex multivariate relationships. Statistical information theory provides a framework for quantifying in a single value the proportion of total information in one set of measures (Y) explained by another set of measures (X). It also quantifies the amount of redundant information and allows for asymmetry of explained information between variables. The general information theoretic approach is presented and illustrated using measures of affect, cognition and behavior. A statistically significant and asymmetric information theoretic relationship is found among the variables: affect (like/dislike) provides a higher percentage of information about behavior (shopping frequency) than behavior does about affect. In addition, affect provides a higher percentage of information about behavior than does perceived location convenience.

Social science researchers have emphasized the need to identify asymmetric relationships among variables (e.g., James, 1981; Miller, 1969, 1975; Miller &

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Farr, 1971; Stewart & Love, 1968; Tyler, 1982). The bimultivariate redundancy statistic discussed by Miller and by Miller and Farr was developed by calculating the proportion of total variance in one set of measures (Y) explained by another set (X). Information theory provides an alternative and non-covariance based approach by calculating the proportion of total information in one set of measures (Y) explained by another set (X). The bimultivariate redundancy statistic is restricted to use with interval level data. In contrast, the information theoretic measure does not assume a linear model, is applicable to nominal level categorical data, and yields an asymptotic distribution known to be chi-squared so that the statistical significance of these relationships can be determined.

The purpose of this article is to present information theory as a method for identifying redundancy (informational overlap) among variables and potential asymmetry between variables. The information theoretic approach is illustrated through example calculations and the article concludes with an empirical analysis of the relationships among affect (like/dislike), cognition (perceived location convenience), and behavior (shopping frequency). The resulting relationships between the variables is shown to be consistent across three distinct stimulus objects (stores) and not idiosyncratic to one stimulus object.

Information Theory and Information Theoretic Measures

Statistical information theory originated in communications theory as a method for quantifying the amount of stochastic uncertainty in a system. Its application has now been extended to numerous other disciplines. For example, in behavioral research information theoretic methods have been applied to brand switching (Bass, 1974; Charnes, Cooper & Learner, 1978; Herniter, 1973, 1974; Sharma, 1981), product vulnerability (Charnes, Cooper, Learner, & Phillips, 1984), discriminant analysis variable selection (Goldstein & Dillon, 1978), analysis of data quality (Golden & Attir, 1979) and analysis of questionnaire data (Brockett, Haaland & Levine, 1981).

Unconditional Entropies

The basic building block of information theory is the entropy functional. It is widely recognized that the calculation of the entropy for a probabilistic system is the standard way to quantify the amount of uncertainty in that probabilistic system (e.g., Gokhale & Kullback, 1978; Guiasu, 1977; Kullback, 1959; Theil & Fiebig, 1984). By way of definition, the entropy of a discrete variable is the sum of the numbers $p \ln(1/p) = -p \ln(p)$ as p ranges over the probabilities of possible values for the variable. Thus, for univariate distributions (p_i), entropy is $H(Y) = -\sum_i p_i \ln(p_i)$. Bivariate and trivariate entropies are analogously defined:

Table 1

Cross-Classification of Responses to "Women's Role" Question from NORC Survey with Demographic Variables

<u>Sex of Respondent</u>	<u>Education of Respondent (Years)</u>	<u>Number Agreeing (Y=1)</u>	<u>Number Disagreeing (Y=2)</u>	<u>Total</u>
Male (X=1)	≤ 8 (Z=1)	72	47	119
	9 - 12 (Z=2)	110	196	306
	≥ 13 (Z=3)	44	179	223
	Total	226	422	648
Female (X=2)	≤ 8 (Z=1)	86	38	124
	9 - 12 (Z=2)	173	283	456
	≥ 13 (Z=3)	28	187	215
	Total	287	508	795

$H(X,Y) = -\sum_i \sum_j p_{i,j} \ln(p_{i,j})$ and $H(X,Y,Z) = -\sum_i \sum_j \sum_k p_{i,j,k} \ln(p_{i,j,k})$ respectively. These quantities measure the uncertainty in the stochastic response distribution of the univariate, bivariate, and trivariate joint distributions, respectively, in units called *bits* (binary information digits) commonly encountered in computer nomenclature.¹

To illustrate the numerical computation of these quantities we examine the results of three questions asked in the 1975 National Opinion Research Center's General Social Survey (National Opinion Research Center, 1975, p. 51). These are: X = sex of the respondent; Y = the answer to 'Do you agree or disagree with this statement? "Women should take care of running their homes and leave running the country up to men."'; and Z = respondent's educational level. Table 1 presents the results from 1443 respondents.

Using the contingency table information from Table 1, we calculate that the univariate distribution of the variable X (sex) is $P(X = 1) = 648/1443 = .4491$, $P(X = 2) = 795/1443 = .5509$, the univariate distribution for Y (agree-disagree) is $P(Y = 1) = 513/1443 = .3555$, $P(Y = 2) = 930/1443 = .64449$, and the univariate distribution for Z (education) is $P(Z = 1) = 243/1443 = .1684$, $P(Z = 2) = 762/1443 = .5281$, $P(Z = 3) = 438/1443 = .3035$. The bivariate distribution of the pair of variables X (sex) and Z (education) is $P(X = 1, Z = 1) = 119/1443 = .0825$, $P(X = 1, Z = 2) = 306/1443 = .2121$, $P(X = 1,$

¹ In fact, the entropy functional can be characterized as the unique way to measure uncertainty subject to certain intuitively reasonable axioms (cf., Guisasu, 1977; Haaland, Brockett & Levine, 1979).

$Z = 3) = 223/1443 = .1545$, $P(X = 2, Z = 1) = 124/1443 = .0859$, $P(X = 2, Z = 2) = 456/1443 = .3160$, $P(X = 2, Z = 3) = 215/1443 = .1490$, the bivariate distribution of the pair of variables X (sex) and Y (agree-disagree) is $P(X = 1, Y = 1) = 226/1443 = .1566$, $P(X = 1, Y = 2) = 422/1443 = .2924$, $P(X = 2, Y = 1) = 287/1443 = .1989$, $P(X = 2, Y = 2) = 508/1443 = .3520$ and the bivariate distribution of the pair of variables Y (agree - disagree) and Z (education) is $P(Y = 1, Z = 1) = 158/1443 = .1095$, $P(Y = 1, Z = 2) = 283/1443 = .1961$, $P(Y = 1, Z = 3) = 72/1443 = .0499$, $P(Y = 2, Z = 1) = 85/1443 = .0589$, $P(Y = 2, Z = 2) = 479/1443 = .3319$, $P(Y = 2, Z = 3) = 366/1443 = .2536$. The trivariate distribution is obtained by dividing each of the entries in the contingency table by the total number of respondents, $P(X = 1, Y = 1, Z = 1) = 72/1443 = .0499$, $P(X = 1, Y = 1, Z = 2) = 47/1443 = .0326$, ..., $P(X = 2, Y = 2, Z = 3) = 187/1443 = .1296$.

Using the formulas previously presented, univariate, bivariate and trivariate entropies can be calculated from the contingency table data. These calculations and results are:

$$\begin{aligned} H(X) &= -.4491\ln(.4491) - .5509\ln(.5509) = .68795, \\ H(Y) &= .3555\ln(.3555) - .6445\ln(.6445) = .65079, \\ H(Z) &= .1684\ln(.1684) - .5281\ln(.5281) - .3035\ln(.3035) = .99907, \\ H(X, Y) &= .1566\ln(.1566) - .2924\ln(.2924) - .1989\ln(.1989) - .3520\ln(.3520) = 1.33866, \\ H(X, Z) &= -.0825\ln(.0825) - .2121\ln(.2121) - \dots - .1490\ln(.1490) = 1.69183, \\ H(Y, Z) &= -.1095\ln(.1095) - .1961\ln(.1961) - \dots - .2536\ln(.2536) = 1.59208, \\ H(X, Y, Z) &= -.0499\ln(.0499) - .0326\ln(.0326) - \dots - .1296\ln(.1296) = 2.27275. \end{aligned}$$

Conditional Entropies

In a multivariate analysis of the informational content of stochastic distributions, it is also of interest to determine the amount of information in one variable's distribution when the various levels of another variable are known. As a step in this direction, *conditional* entropies are used to quantify the amount of *uncertainty remaining* in the distribution of a variable Y after some other variable X has been observed. The conditional entropies are the ordinary (simple) entropies calculated using the conditional probabilities.

Conditional entropy formulae are given by $H(Y|X) = \sum_j H(Y|X = j)P(X = j)$ where $H(Y|X = j) = -\sum_i p_{ij}\ln(p_{ij})$, and where the conditional probabilities are defined in the usual manner via $p_{ij} = P(Y = i|X = j) = p_{ij} / \sum_i p_{ij}$. Thus, the conditional entropy $H(Y|X)$ is the average of the uncertainty in the Y distribution (conditional entropy) calculated for each of the possible specific values $X = j$. It can be shown that it satisfies $0 \leq H(Y|X) \leq H(Y)$.

The bivariate and trivariate conditional entropies can also be determined without actually calculating each of the individual conditional probabilities. This is done by using the univariate and bivariate entropies via the formula $H(Y|X) = H(X, Y) - H(X)$. For example, using the National Opinion Research Center's (NORC) data on women's roles from Table 1 and the unconditional entropies presented earlier, $H(Y|X) = 1.33866 - .68795 = .65071$. $H(Y|Z)$ and $H(X|Z)$ would be calculated similarly. By the same logic and algebraic techniques (cf., Guiasu, 1977), it can be shown that the amount of uncertainty remaining in Y after observing both of the variables X and Z (trivariate conditional entropy) is given by the formula $H(Y|X, Z) = H(X, Y, Z) - H(X, Z)$, which, for the NORC data is $H(Y|X, Z) = 2.27275 - 1.69183 = .58092$.

Amount of Information

The *amount of shared or redundant information* that one variable X supplies about another variable Y can be quantified by the *difference*, $I(Y|X)$, between the amount of uncertainty which we had about Y just before and just after having observed X : $I(Y|X) = H(Y) - H(Y|X)$. Hence, information theory defines information to be the *reduction in uncertainty* in one variable when another variable is known. Any reduction in uncertainty (namely the quantity $H(Y) - H(Y|X)$) is acknowledged to be due to information furnished or explained by X about Y . In a similar manner, the quantity $I(Y|X, Z) = H(Y) - H(Y|X, Z)$ measures the amount of information in Y which is supplied by the *pair* of variables X and Z taken together.² For example, for the NORC data presented earlier there is virtually no information about the respondent's view on a woman's role (Y) supplied by knowing the respondent's sex (X) because $I(Y|X) = H(Y) - H(Y|X) = .65079 - .65071 = .00008$. By comparison, the amount of information about the respondent's view on a woman's role (Y) which is supplied by knowing both the respondent's sex (X) and educational level (Z) is $I(Y|X, Z) = H(Y) - H(Y|X, Z) = .65079 - .58092 = .06987$.

It is also of interest in multivariate analysis to measure the relationship between two variables Y and Z when the value of a third variable X is known. In this regard, we shall call the quantity $I[(Y|Z)||X] = H(Y|X) - H(Y|X, Z)$ the *incremental conditional* (on X) information about Y furnished by Z when X is known. Note that this quantity measures the *additional increment in information*

² Note that the inequality $0 \leq I(Y|X) \leq H(Y)$ holds. The left hand inequality becomes an equality when $H(Y|X) = H(Y)$, that is, when there is just as much uncertainty about Y after observing X as there was before observing X . In this case X and Y are independent and furnish no information about each other. The right hand inequality becomes an equality when $H(Y|X) = 0$, that is, when there is no uncertainty in Y after having observed X . In this case Y is completely deterministically dependent on X and the knowledge of X gives maximal (namely all) information about Y .

gained about Y by observing Z after X has been observed (i.e., is already known) and is found by taking the difference between the amount of uncertainty in Y 's conditional distribution before and after knowing Z , keeping in mind all the time that the value X is known.

Most people think in terms of percentages rather than bytes of information (the units of measurement of information) and, hence, for explanatory and interpretative purposes it is useful to put the information furnished by one variable about another on a 0-100 (percentage) scale. The *percentage of information* in Y explained by X is defined by taking the reduction in uncertainty in Y 's distribution which is attributable to having a knowledge of X as a percentage of the original (unconditional) uncertainty amount $H(Y)$. Mathematically this percentage of information in Y explained by X is defined as $100[I(Y|X)/H(Y)] = 100\{[H(Y) - H(Y|X)]/H(Y)\}$.

The concept of percent of information and R^2 (in regression) are similar in that both can be viewed as attempting to assess dependency relationships as a function of the amount of uncertainty in one variable reduced by measurement of other variables. There are, however, some marked differences between R^2 and the information theoretic percentage of information measure. These are discussed in Appendix 1.

Statistical Significance

The information theoretic framework allows for statistical testing of the strength of relationships on a continuum from complete independence (corresponding to zero information furnished) to complete determinacy (which may be a non-linear relationship corresponding to the maximal information). If X and Y are independent (so that the true value of $I(X|Y)$ is zero) and if the probability distributions for X and Y arise from a sample of size n , then the empirically calculated value of $I(Y|X)$ converges to zero at a rate of $1/n$. It can be shown under these conditions $2nI(Y|X)$ has (asymptotically) a χ^2 distribution with degrees of freedom equal to $(k_X - 1)(k_Y - 1)$ where k_X and k_Y are the number of response categories (answers) available for X and Y respectively.

The above formula can be used to test the null hypothesis of independence of X and Y , or test equivalently that there is zero redundant or shared information between X and Y (Gokhale & Kullback, 1978; Kullback, 1959). When there is statistical significance, X furnishes a significant amount of information about Y . For example, using the NORC data presented earlier, the amount of information furnished about a respondent's view on a woman's role (Y) by knowing the respondent's sex (X) was $I(Y|X) = .00008$, so that $2nI(Y|X) = 2 \times 1443 \times .00008 = .23088$. This is statistically insignificantly different from zero using the χ^2 distribution with $(2 - 1)(2 - 1) = 1$ degree of freedom. In a similar manner, the test statistic $2nI(Y|X, Z)$ has $(k_Y - 1)(k_X k_Z - 1)$ degrees

of freedom, and the incremental conditional information test statistic $2nI[(Y|Z)||X]$ has $k_X(k_Y - 1)(k_Z - 1)$ degrees of freedom.³ The critical values for the chi-squared distribution with v degrees of freedom needed to perform these tests can be extremely well approximated, even for fairly small values of v , by the Wilson - Hilferty hand calculation formula $\chi^2_{v,\alpha} = v[1 - (2/9v) + Z_\alpha(2/9v)^{1/2}]^3$ where $\chi^2_{v,\alpha}$ is the α^{th} percentile of the χ^2_v distribution and Z_α is the α^{th} percentile of the standard normal distribution (cf., Zar, 1978).

Research Questions and Contribution

Having presented the theory and simple illustrative example showing the calculation of the entropic measures, this section develops a substantive application of the techniques in an investigation of relationships among consumer behavior variables. Each of the research questions examined is stated in terms of the percentage of information furnished by one variable or set of variables about another variable. Formally, the percentage of information in X_1 about Y is $100[I(Y|X_1)/H(Y)]$, while the percentage of information the pair of variables X_1 and X_2 gives about Y is denoted by $100[I(Y|X_1, X_2)/H(Y)]$. This study investigates the following research questions.

1. For any designation of the pair of variables, X and Y , is $100[I(Y|X)/H(Y)] > 0$? Is the relationship between pairs of variables, X and Y , symmetric or asymmetric?

2. For any designation of the three variables X_1 , X_2 and Y is $100[I(Y|X_1, X_2)/H(Y)] > 0$?

3. For any designation of the three variables X_1 , X_2 and Y is $100\{I[(Y|X_2)||X_1]/H(Y|X_1)\} = 100\{[H(Y|X_1) - H(Y|X_1, X_2)]/H(Y|X_1)\} > 0$?

The first research question addresses the percent of bivariate redundant (shared) information between variables X and Y by asking if X provides a significant percent of information about Y and by changing the designation of X and Y , the symmetry or asymmetry between variable pairs. If only one directional relationship is postulated, then only a single percentage of information $100[I(Y|X)/H(Y)]$ needs to be calculated (i.e., the reverse designation of X and Y would not be tested).

³ The test statistic $2nI(Y|X)$ derived here is mathematically equivalent to the minimum discrimination information (MDI) statistic based on constrained Kullback - Leibler information numbers (cf., Gokhale & Kullback, 1978). In fact, the quantity $I(Y|X)$ is precisely the Kullback - Leibler information number for distinguishing between the joint probability measure for (X, Y) and the product measure for X and Y . The test statistic $2nI(Y|X)$ is also equivalent to the log likelihood ratio statistic based on loglinear contingency table models for testing these same hypotheses (cf., Haberman 1978). The mathematical development leading to the MDI, loglinear log likelihood ratio, and difference of uncertainty test statistics are, however, quite different.

The second research question considers the trivariate redundant (shared) information among three variables (X_1 , X_2 and Y) and asks if the pair of variables X_1 and X_2 provides a statistically significant percent of information about Y . Again, if directionality of information flows is postulated a priori, then only a subset of all possible permutations of percentage of information triads need be examined. Finally, the third research question addresses whether or not there is a statistically significant amount of incremental conditional information gained about Y by observing X_2 when X_1 is known.

In this study the information theoretic research questions are used to investigate the relationships among affect, cognition and behavior (using measures of like/dislike, perceived location convenience and patronage frequency for three retail store chains). Although a number of other studies have focused on relationships between affect, cognition and behavior as related to attitude (e.g., Bagozzi, 1978, 1981; Fishbein & Ajzen, 1975; Reddy & LaBarbera, 1985; cf., Bettman, 1986), no published research has used an information theoretic method to study these relationships.

This study is unique in that it provides insight into the shared information and potential asymmetry between attitudinal components and introduces a method not previously described in the behavioral science literature. In addition, the information theoretic method used in this research goes beyond the asymmetry information provided by the bimultivariate redundancy statistic in that: (a) it is not linear model based; (b) it is applicable to both nominal and interval level data; and (c) the asymptotic distribution for determining significance of relationships is known for the information theoretic measures (it is chi-squared distributed) whereas for the bimultivariate redundancy statistic the distribution is only conjectured to be approximately F distributed based on Monte Carlo simulations (Miller, 1975).

Method

A sample of 1600 adults was selected from a nationwide consumer mail panel to represent region, population density and demographic characteristics proportionate to the population of the United States. Because panel members were predominantly female, half of the cover letters asked the panel member to fill out the questionnaire and the other half told the panel member to ask their spouse to respond.

Survey Instrument

Sears, K-Mart and Wards are widely-known nationwide stores with broad-based product lines representing distinctly different market situations, appeals

and strategies. Patronage frequency was elicited categorically for each of the three stores ("How frequently do you go to Sears, K-Mart and Wards?"). Response options represented Never (1), Once a year or less (2), Two to six times a year (3), Seven to twelve times a year (4), Between 12 and 52 times a year (5), and Once a week or more often (6).

At the time of mailing, respondents were randomly assigned to a graphic or numerical scale format for affect and location convenience measures. The numerical scale required that a number be written from a bi-polar scale (1 - 7) in a space provided for each store. Graphic treatment respondents wrote the first letter for each store (S, K or W) above the place on a calibrated line that represented their answer (between bi-polar adjectives). From left to right, affect adjectives were "dislike" - "like" and location adjectives were "inconvenient location" - "convenient location".

Final Sample and Preliminary Analyses

The final sample consisted of 894 usable surveys. Using the chi-squared test, there were no significant differences in response rates, demographics or patronage frequency between numerical ($n = 453$) and graphic ($n = 441$) scales. In addition, when graphic responses were coded in a manner consistent with the numerical response options (1, 2, 3, ..., 7), the chi-squared test showed no significant distributional differences between the two formats for affect or location convenience for any store. Thus, the remainder of the analyses combine the data from the two formats.

Respondents were 41.3 percent male, slightly over 50 percent of the respondents had some college or higher (39.4 percent of the sample graduated high school only), approximately two-thirds of the respondents earned less than \$30,000 a year, and approximately 55 percent were under 50 years old. Only respondents answering *all* affect, location and behavior questions were included in illustrative analyses presented below.

Information Theoretic Analyses and Results

Based on the formulae in the information theory section, a complete entropic taxonomy of the joint multivariate distribution of affect, location and frequency was developed. Initially, the univariate, bivariate and trivariate entropies were calculated. These unconditional entropies shown in Table 2 represent the actual amount of uncertainty in each of the observed variables. Comparison of the unconditional entropies with the maximum entropies corresponding to purely random distributions shows how much information was gained (or, conversely, how much uncertainty was reduced) by observing the variable(s). That is, the

Table 2
Summary of Unconditional Information Theoretic Results

<u>Unconditional Entropy</u>	<u>Sears</u>	<u>K-Mart</u>	<u>Wards</u>
$H(\text{Affect})^a$	1.88	1.84	1.88
$H(\text{Location})^a$	1.83	1.65	1.84
$H(\text{Frequency})^b$	1.50	1.64	1.55
$H(\text{Affect, Location})^c$	3.63	3.42	3.65
$H(\text{Affect, Frequency})^d$	3.23	3.28	3.21
$H(\text{Location, Frequency})^d$	3.26	3.14	3.26
$H(\text{Affect, Location, Frequency})^e$	4.84	4.65	4.77
<u>Percent Deviation from Purely Random^f</u>	<u>Sears</u>	<u>K-Mart</u>	<u>Wards</u>
$\ln H(\text{Affect})$	3.51	5.61	3.15
$\ln H(\text{Location})$	6.20	15.12	5.22
$\ln H(\text{Frequency})$	16.00	8.67	13.62
$\ln H(\text{Affect, Location})$	6.68	12.09	6.22
$\ln H(\text{Affect, Frequency})$	13.53	12.29	14.04
$\ln H(\text{Location, Frequency})$	12.88	16.02	12.71
$\ln H(\text{Affect, Location, Frequency})$	14.90	18.15	16.11

^a Maximum entropy = $\ln 7 = 1.95$. ^b Maximum entropy = $\ln 6 = 1.79$. ^c Maximum entropy = $\ln 49 = 3.89$. ^d Maximum entropy = $\ln 42 = 3.74$. ^e Maximum entropy = $\ln 294 = 5.68$. ^f Calculated by $100[1 - H(X)/\text{Maximum Entropy}]$; 0% is purely random and 100% is purely deterministic.

bottom section of Table 2 shows how much each unconditional entropy deviates from a purely random distribution (maximum entropy). For example, Table 2 illustrates that for Sears patronage frequency has the largest percent deviation from a purely random distribution, whereas for K-Mart and Wards the joint distribution of affect, location and frequency has the highest deviation from a purely random distribution.

The unconditional entropies shown in Table 2 are used as a baseline for judging first the amount of uncertainty in the distributions of the individual variables, and then the amount of uncertainty in the joint distributions of variables. In essence, these unconditional entropies represent a priori levels of uncertainty in the variables which can be compared with the levels of uncertainty *after* observing a new piece of information (i.e., another variable). Thus, the unconditional entropies were used as a building block in the calculations conducted to address the research questions.

Table 3
Summary of Conditional Information Theoretic Results

<u>Conditional Entropy</u>	<u>Sears</u>	<u>K-Mart</u>	<u>Wards</u>
$H(\text{Affect} \text{Location})^a$	1.81	1.77	1.81
$H(\text{Location} \text{Affect})^a$	1.75	1.58	1.77
$H(\text{Affect} \text{Frequency})^a$	1.73	1.64	1.67
$H(\text{Frequency} \text{Affect})^b$	1.35	1.44	1.33
$H(\text{Location} \text{Frequency})^a$	1.75	1.50	1.71
$H(\text{Frequency} \text{Location})^b$	1.43	1.49	1.42
$H(\text{Affect} \text{Location, Frequency})^a$	1.58	1.51	1.51
$H(\text{Location} \text{Affect, Frequency})^a$	1.60	1.37	1.55
$H(\text{Frequency} \text{Affect, Location})^b$	1.20	1.23	1.12
<u>Percent of Information^c</u>	<u>Sears</u>	<u>K-Mart</u>	<u>Wards</u>
In affect given by location	3.78	3.65	4.21
In location given by affect	3.89	4.06	4.30
In affect given by frequency	8.02	10.60	11.63
In frequency given by affect	10.01	11.90	14.17
In location given by frequency	4.05	9.03	7.01
In frequency given by location	4.91	9.11	8.36
In affect given by location and frequency	15.82	17.62	20.12
In location given by affect and frequency	12.07	16.83	15.69
In frequency given by affect and location	19.93	24.78	27.74

^a Maximum entropy = $\ln 7 = 1.95$. ^b Maximum entropy = $\ln 6 = 1.79$. ^c Calculated by $100[1 - H(X|Y)/H(X)]$; if 0%, then knowing Y does not change the uncertainty in X ; if 100%, then Y completely determines X . All percent of information significant within $p < .01$ for 36 d.f. when affect and location is considered, otherwise d.f. = 30.

Bivariate Redundancy and Symmetry between Pairs of Variables

The first research question asked if X provides a statistically significant amount of information about Y and queries the resultant symmetry or asymmetry. This question was addressed by using the unconditional entropies from Table 2 to calculate the conditional entropies, and then for determining the percent of information that each variable shares with each other variable. These results are presented in Table 3. The conditional entropy can be thought of as the *posterior* uncertainty remaining in one variable after the observation of another variable.

It can be judged relative to the unconditional entropy (*prior* uncertainty) of the variable being scrutinized.

Because there is no reason to assume directionality in the relationships *a priori*, all possible designations of variables (X, Y) are examined. Behavioral theory suggests that attitudes (e.g., affect and location convenience perception) can be postulated to either precede or follow behavior. Classical attitudinal theory postulated significant association between attitudes and behavior to suggest that attitudes *caused* behavior until Festinger (1957) and Bem (1967) suggested the reverse (behavior $\rightarrow$ attitude). Thus, we are interested in investigating potential asymmetry between attitudinal perceptions and behavior. In terms of the operationalizations herein, behavior (shopping frequency) may influence perception and, in turn, perception may impact how much a person shops at the store. Information theory allows us to study whether this influence is symmetrical or asymmetrical.

The results indicate that there is a statistically significant amount of redundancy in the information content between *all* variable pairs. In addition, inspection of Table 3 shows that the information relationships between variables are asymmetric. For example, affect explains more information about frequency than frequency explains about affect. For Sears, affect explains 10.01 percent of the information in frequency while frequency only explains 8.02 percent of the information in affect. This pattern of asymmetry is consistent across three store chains. Thus, these results suggest that observing affect, for example, provides a larger percentage of information about patronage frequency than observing patronage frequency does about affect. In addition, the exact magnitude of information revealed about patronage frequency by affect and by location varies across stores, but it is important to note that affect consistently provides a higher percentage of information about behavior than does location convenience.

Trivariate Redundancy among Variables

The second research question asked if there were a statistically significant percent of information in any combination of two variables about the third member of the triad. The last three rows in Table 3 provide the results related to this question. The answer is affirmative for all cases, as all permutations of results were statistically significant.

As shown in Table 3 for all three stores, the largest percentage of shared information is provided by frequency when affect and location are known. Together affect and location yield approximately 20 percent (for Sears) to 28 percent (for Wards) of the information in patronage frequency. These results suggest that uncertainty about patronage frequency can be reduced with information about consumer perceptions of affect and location convenience. In

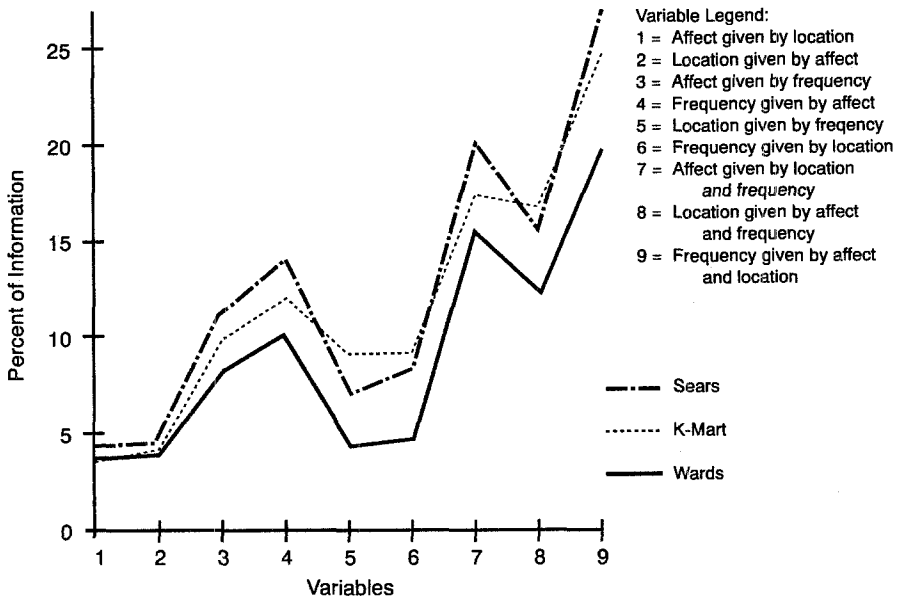


Figure 1

Bivariate and Trivariate Percent Redundant Information for Each Store

addition, location and frequency provide more information about affect than do affect and frequency about location.

The relative magnitudes of information furnished in the bi- and trivariate cases are consistent across stores. This is portrayed graphically in Figure 1. The information relationships among the variables are not merely idiosyncratic to each store but, rather, seem to be indicative of general characteristics of consumer choice processes (at least as investigated herein).

Incremental Conditional Information among Variables

After a single variable has been observed for the purpose of providing information about a second variable, the question arises about the value of observing yet a third variable for the purpose of providing even more information about this second variable. This is the issue addressed by the third research question: Is there a statistically significant increment in the conditional information about a variable Y given by observing a variable X_2 when we already have knowledge of the variable X_1 ? This question was addressed by computing the incremental conditional information for all permutations of the three variables. The results of these computations are provided in Table 4.

Table 4

Summary of Incremental Conditional Information Theoretic Results

Additional Information Given By One Variable about Another When A Third is Known	Sears	K-Mart	Wards
$H(\text{Affect} \text{Frequency}) - H(\text{Affect} \text{Location}, \text{Frequency})$	0.15	0.13	0.16
$H(\text{Location} \text{Frequency}) - H(\text{Location} \text{Affect}, \text{Frequency})$	0.15	0.13	0.16
$H(\text{Affect} \text{Location}) - H(\text{Affect} \text{Location}, \text{Frequency})$	0.23 ^a	0.26 ^a	0.30 ^a
$H(\text{Frequency} \text{Location}) - H(\text{Frequency} \text{Affect}, \text{Location})$	0.23 ^a	0.26 ^a	0.30 ^a
$H(\text{Frequency} \text{Affect}) - H(\text{Frequency} \text{Affect}, \text{Location})$	0.15 ^a	0.21 ^a	0.21 ^a
$H(\text{Location} \text{Affect}) - H(\text{Location} \text{Affect}, \text{Frequency})$	0.15 ^a	0.21 ^a	0.21 ^a
% Additional Information Given By One Variable About Another When A Third Is Known	Sears	K-Mart	Wards
Percent Of Information In Affect Given By Location When Frequency is Known	8.67	7.93	9.58
Percent Of Information In Location Given By Affect When Frequency is Known	8.57	14.00	12.28
Percent Of Information In Affect Given By Frequency When Location is Known	12.71 ^a	14.69 ^a	16.57 ^a
Percent Of Information In Frequency Given By Affect When Location is Known	16.08 ^a	17.45 ^a	21.13 ^a
Percent Of Information In Frequency Given By Location When Affect is Known	11.11 ^a	14.83 ^a	15.79 ^a
Percent Of Information In Location Given By Frequency When Affect is Known	8.57 ^a	13.29 ^a	11.86 ^a

^a $p < .01$; χ^2 d.f. = 216, if Patronage Frequency is known, 210 if Affect or Perceived Location Convenience is known; a normal approximation is used to compute significance.

Four of the six incremental conditional information theoretic relationships shown in Table 4 are statistically significant. Moreover, these results are consistent across the three stores. For example, there is a statistically significant percent of incremental conditional information provided by affect about behavior when location convenience perceptions are known. If location perceptions are known, then between 16.08 and 21.13 percent (for Sears and Wards, respectively) more information about frequency is provided by subsequently observing affect. The other three significant mathematical decompositions of incremental

conditional information can be interpreted similarly and suggest that there is a significant amount of statistical dependence among the triad of variables when location or affect is known.

Regarding the statistically insignificant relationships, the results suggest that affect and location are conditionally independent and provide little or no information about each other when patronage frequency is known. Thus, the affirmative answer to the research question is contingent upon which variable is observed. According to these results, affect and location do not provide significant additional information about each other when patronage frequency is known, but frequency and affect, as well as location and frequency, do provide a significant percent of incremental information about each other when the remaining variable in the triad is observed (either location or affect). Thus, based upon the entropic considerations, information theory can provide directions regarding the incremental value of additional variable observations.

Conclusions and Implications

This article presented information theory as a method for identifying redundancy (information overlap) and potential asymmetry in the relationships between variables. The discrete entropic-based analysis used in this article to characterize uncertainty uses only the nominal characteristics of the data and considers the entire multivariate distribution of responses. In addition, the statistical significance of the relationships identified by the information theoretic approach can be tested using a chi square distribution. Thus, the information theoretic procedure is distribution free and, hence, can be used to assess non-linear as well as asymmetric relationships in data. On the other hand, the ordinal scale of measure of some data sets presents other *information* of a non-entropic nature (namely the ordinality) which is not utilized in the purely nominal entropy measures outlined here. Hence, it may be useful to score such data and use metric procedures from classical statistics (such as using regression analysis when the goal of the analysis is to *predict* dependent variables). Generalizations of the procedure outlined here (including logit analysis) can handle ordinal variables and are discussed in Appendix 2.

Several types of information theoretic measures were presented in this article. First, the unconditional entropies, $H(Y)$, $H(X,Y)$, and $H(X,Y,Z)$, assess the amount of uncertainty intrinsic in the observation of the individual variables or sets of variables. These unconditional entropies also serve as a baseline for judging the information gained (uncertainty reduced) by observing other variables. Second, conditional entropies, $H(Y|X)$, measure the residual amount of uncertainty in Y after observing X . The difference between the unconditional and conditional entropies is a measure of shared or redundant information about Y in X . Third,

by calculating the difference between the uncertainty in Y when one variable is known, $H(Y|X)$, and the uncertainty in Y when two variables are known, $H(Y|X, Z)$, we have a measure of the incremental conditional information in Y furnished by Z beyond the information furnished by X . These measures of information can be compared to the original baseline uncertainty (entropy) in order to obtain a (asymmetric) measure of the percent of information one variable supplies about another.

Information theoretic analysis was applied to data obtained from a cross-sectional national sample of 894 subjects who responded to questions about their affect (like/dislike), cognition (perceived location convenience) and behavior (patronage frequency) for three different stimuli (stores). The results of the information theoretic analyses applied to the data illustrate the use of the various measures of entropy and provide insight into the relationships among affect, cognition, and behavior.

The information theoretic analyses in this article suggest that affect, perceived location convenience and patronage frequency behavior are related variables and that the information content between pairs of these variables is asymmetric. In addition, the relationships among affect, patronage frequency and perceived location convenience are complex. The nature of the asymmetry between variables indicates that affect provides a higher percentage of information about patronage frequency than patronage frequency provides about affect. These results suggest that from an information theoretic perspective, while affect and behavior do have a significant amount of shared information, they also have distinct information components.

Perceived location convenience also provides information about behavior but does not provide as much information about behavior as does affect. Together affect and perceived location convenience can provide substantial information about behavior. In addition, perceived location convenience provides a higher percentage of information about behavior than it provides about affect. This implies that cognitions about location may function through behavior to influence affect or that a convenient location may influence patronage but not necessarily affect toward to the store. This is further supported by an important finding of the study: Once behavior is known, affect and perceived location convenience are independent of each other (i.e., there is no statistically significant amount of incremental information about affect furnished by location beyond the information already furnished by behavior).

In summary, this article has illustrated that information theoretic methods provide insight into the identification of shared information and asymmetric relationships among variables. Information theoretic methods are valuable in that they provide a measure of the strength of a relationship between scaled variables. Moreover, the examination of the shared information among pairs of

variables and the incremental conditional information available in the multivariate structure was not idiosyncratic to each store, but rather was consistent across the three retail outlets.

The concept of asymmetry between variables is not new. Behavioral researchers have previously identified the need for measuring information redundancy and asymmetry (cf., James, 1981). This article addresses that need by presenting information theory as a viable methodological alternative to metric measures of redundancy and asymmetry, such as the bivariate redundancy statistic (Miller & Farr, 1971). Social science researchers can utilize information theory to better understand conceptual and theoretical relationships when metric models are not appropriate. In addition, when an information theoretic goal is identified, entropy theory can be useful in identifying the variables most fruitful to measure as predictors of other variables.

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Appendix 1

Comparison of Information Theoretic and Classical Linear Model Approaches

Although there are similarities in conceptual purpose, there are several differences between the information theoretic *percentage of information* contained in one variable (or variables) about another described in this article (i.e., explained information) and the *explained variance* measure (R^2) used in classical linear models. Some of these are enumerated below:

1. Explained variance represents the proportion of *variation* in Y explained by X and its calculation requires interval scaled numerical data. In behavioral research, however, it is common to collect data using categorical measures. Hence, in order to use variance measures the researcher must make the choice of a numerical scoring method. This choice can be important because the scoring method used can affect which variables are significant in a regression (Golden & Brockett, 1987). Moreover, in some situations, the categorical measures under investigation involve nominal variables (such as race, religion, country, primary language, etc.), which cause other problems in using the linear model techniques.

The entropic information measure offers an advantage in that it does not require numerical assignment of scores to categories or even require ordinal data, because its calculation and interpretation involves only the *probabilities* of the states that the variable(s) may obtain and not the numerical values themselves. Thus, standard assumptions of normality and linearity are not required for calculation and interpretation of information theoretic analysis. On the other hand, the linear model approach applied to interval level data allows the analyst to use the numerical values of the dependent variables in order to predict the numerical value of the independent variable, a capability not possessed by the entropic approach. Thus, significant relationships can be determined by the entropic approach but prediction requires another methodology. Extensions of the information theoretic technique capable of incorporating ordinal variables are given in Appendix 2.

Whereas entropic and information theoretic analysis can be applied to continuous variables, it is not as convenient as regression for continuous variables with a wide range of values, and grouping can result in information loss. However, when discrete categories are used, and especially when the data are nominal, information theoretic analysis can provide useful insights into the stochastic structure.

2. The explained variance measure is unity if and only if the variables X and Y are *linearly related*. Nonlinear structural relations between X and Y may not

		Y				
		1	2	3	4	5
X	1					X
	2		X			
	3			X		
	4				X	
	5	X				

Figure 2

Two Variables Whose Relationship is Unrecognized by Regression ($R^2 = 0$), but Recognized by Information Theory Methods [$H(X|Y) = 0$].

be reflected in explained variance. On the other hand, the entropic explained information measure is unity if and only if X and Y have a *deterministic relationship* and both linear and non-linear relationships can be identified.

3. Explained variance is zero if and only if the best fitting straight line relationship between X and Y is a horizontal line; however, variables can easily have this property and still be highly and even deterministically related (e.g., take X symmetrically distributed about zero and $Y=X^2$). On the other hand, the entropic explained information measure is zero if and only if X and Y are *independent*. Thus, a low value for the information theoretic measure is more indicative of a low level of statistical dependence than is a low value of explained variance in a linear model.

Figure 2 portrays a situation in which there is a nonlinear relationship between two variables X and Y . In this Figure the marginal distributions for both X and Y exhibit maximal uncertainty (entropy) and a regression of X on Y or a regression of Y on X leads to an R^2 of 0, yet X and Y are deterministically related. The information theoretic method recognizes this deterministic relationship between the variables, because one can calculate $H(X|Y) = 0$, and hence determine that the percentage of explained information is 100%.

Appendix 2

Comparison of Information Theoretic and Other Discrete Data Model Approaches

The uncertainty (entropic) based statistics presented here are particular cases of general constrained information theoretic estimation techniques based upon the so-called Kullback-Leibler numbers. This more general information theoretic technique also contains the methodologies of loglinear contingency table modeling, and certain logit and probit models as well. Although the information theoretic test statistics developed in this article and the loglinear contingency table models lead to identical conclusions about global dependency structure, the decompositions used in the entropic information theoretic approach are interpretable as the expected amounts of information in one variable (or set of variables) explained by another variable (or set of variables). In the case of the marginal conditional information, the entropic information theoretic approach determines the expected amount of information one variable contains about another variable when the level of a third variable is fixed. If there is a significant amount of shared information, then the conditional independence hypotheses are rejected. By contrast, in the usual loglinear approach these same hypotheses do not have an immediate information interpretation because these hypotheses are evaluated by testing if certain hyper-parameters in a multiplicative probability structure model are identically zero.

To develop the loglinear model from an information theoretic perspective we start by delineating the minimal set of parameters needed in conjunction with hypotheses to be tested in order to completely describe the probability structure. For example, this would be the marginal probabilities of the two variables in the independence model because independence together with marginal probabilities completely specifies the contingency table structure in this case. When testing whether or not there is a significant amount of marginal incremental information, this minimal set of parameters includes the conditional probabilities for the two variables given the level of a third variable. The parameters in the minimal set are then constrained to equal the observed values of the parameters in the sample.

In general, these constraints alone do not completely determine the structure without the hypothesis to be tested, and there may be many models which satisfy these few constraints. Accordingly, from among the class of all probability distributions which preserve the desired observed structure, the model we select is the most uncertain (maximum entropy) distribution. Formulated mathematically the problem becomes to pick $\mathbf{p} = (p_i)$ to solve the programming problem:

$$(A.1.) \quad \begin{aligned} \text{Max } H(\mathbf{p}) &= -\sum p_i \ln(p_i) \\ \mathbf{A}\mathbf{p} &= \mathbf{b} \\ \mathbf{p} &\geq 0 \end{aligned}$$

For notational convenience here we have ordered the contingency table cell probabilities lexicographically and renumbered them $p_1, p_2, p_3, \dots$. The first row of the matrix $\mathbf{A}$ is selected to be all 1s and b_1 is selected to be 1 so that the first constraint is $\sum p_i = 1$.

By introducing a Lagrange multiplier for each of the constraints in Equation A.1., it can be shown (cf., Brockett, Charnes, & Cooper, 1980; Gokhale & Kullback, 1978) that the optimum values are of the form $p_i^* = \exp(\mathbf{z}_i \mathbf{A}_i)$ where $\mathbf{A}_i$ denotes the i^{th} column of the matrix $\mathbf{A}$ and $\mathbf{z} = (z_i)$ is the collection of Lagrange multipliers. In the case in which the entries of the matrix $\mathbf{A}$ are all 0 or 1, the constraint Equation A.1. specifies that certain designated marginal probabilities are equal to $\mathbf{b}$. The vector $\mathbf{b}$ is usually chosen to be the corresponding set of *observed* marginal probabilities. In this case, the information theoretic fit $p_i^* = \exp(\mathbf{z}_i \mathbf{A}_i)$ is just the loglinear or multiplicative model for the contingency table probabilities, and thus loglinear modeling is a particular case of the information theoretic approach, but one in which the parameterization is determined from a fundamental approach. Moreover, in the above entropic development, testing whether certain constraints are superfluous is equivalent to testing whether the corresponding Lagrange multiplier parameter $\mathbf{z}$ is zero in the loglinear model.

For the hypotheses considered in this article, testing whether these Lagrange multipliers are zero leads to the same test statistic developed from non-extremal principles in this article (e.g., $I(Y|X) = H(Y) - H(Y|X)$, or $I(Y|X, Z) = H(Y) - H(Y|X, Z)$, or $I[(Y|Z)|X] = H(Y|X) - H(Y|X, Z)$) so that the loglinear approach follows from the information theoretic approach. See Gokhale and Kullback (1978), Brockett et al. (1980), or Charnes, Cooper, Learner and Phillips (1980) for more details. Gokhale and Kullback (1978) give the pertinent covariance matrices for inference.

The Logit model is a special case of the loglinear model (see Green, Carmone, and Wachspress, 1977, p. 56) and, hence, can also be derived as a particular case of the information theoretic technique. In the logistic model the log-odds ratio in favor of occurrence of a particular event E is related to several independent variables $x_1, x_2, x_3, \dots$, via the relationship $\ln\{P(E|x)/[1 - P(E|x)]\} = \sum_i \beta_i x_i$, where the β_i s are parameters to be fit to the observed data. To illustrate that the constrained information theoretic generalization of the entropic analysis presented here also contains the Logit model for contingency table analysis, we reproduce the example in Charnes et al. (1980) which uses the following four samples under different values of a single covariate x :

<u>Sample #</u>	<u>Value of x</u>	<u>Sample Size</u>	<u># of Successes</u>
1	0	10	1
2	1	10	6
3	2	10	3
4	3	10	
8			

These are transformed into a contingency table format:

<u>Value of x</u>	<u>i</u>	<u>Success (j = 1)</u>	<u>Failure (j = 2)</u>	<u>Total</u>
0	1	1	9	10
1	2	6	4	10
2	3	3	7	10
3	4	<u>8</u>	<u>2</u>	<u>10</u>
	Total	18	22	40

If we solve the maximum entropy problem below,

$$\begin{aligned}
 \text{Max } H(\mathbf{p}) = & - \sum_{i=1}^4 \sum_{j=1}^2 p_{ij} \ln(p_{ij}) \quad \text{subject to} \\
 & \sum_{j=1}^2 p_{ij} = \frac{10}{40} \quad i = 1, 2, 3, 4 \\
 & \sum_{i=1}^4 p_{i1} = \frac{18}{40} \\
 & \sum_{i=1}^4 x_i p_{i1} = \frac{36}{40} \\
 & p_{ij} \geq 0 \quad i = 1, 2, 3, 4; j = 1, 2
 \end{aligned}
 \tag{A.2}$$

then in terms of the formulation Equation A.1., the matrix constraint $\mathbf{Ap} = \mathbf{b}$ is of the form

$$\begin{vmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ x_1 & 0 & x_2 & 0 & x_3 & 0 & x_4 & 0 \end{vmatrix} = \begin{vmatrix} p_{11} \\ p_{12} \\ p_{21} \\ p_{22} \\ p_{31} \\ p_{32} \\ p_{41} \\ p_{42} \end{vmatrix} = \begin{vmatrix} 1 \\ 10/40 \\ 10/40 \\ 10/40 \\ 10/40 \\ 18/40 \\ 36/40 \end{vmatrix}$$

From the loglinear representation of the optimal solution in terms of the columns of the matrix A and the Lagrange multipliers, we find

$$\ln(p_{11}) = x_1 z_7 + z_6 + z_2 + z_1 \text{ and } \ln(1 - p_{11}) = \ln(p_{12}) = z_2 + z_1,$$

so that

$$\ln[p_{11}/(1 - p_{11})] = x_1 z_7 + z_6.$$

Similarly, for $i = 2, 3, 4$, after calculating the loglinear representation for p_{i1} and $1 - p_{i1} = p_{i2}$ we find the same coefficients as above so that $\ln[p_{i1}/1 - p_{i1}] = x_i z_7 + z_6$. This is the logistic model $\ln[P(E|x)/1 - P(E|x)] = \beta_0 + \beta_1 x$ with the Lagrange multipliers serving the role of the logistic parameters. Thus, logistic modeling can also be developed from the perspective of the entropic methods discussed here.

Another technique which is often used for the analysis of categorical data like that examined in this article is Probit analysis. This also can be motivated from a general information theoretic approach. For details of this development see Charnes et al. (1980).

In summary, several of the alternative methods of discrete data analysis can be derived as special cases of the information theoretic approach. Some of these such as logit analysis allow the incorporation of ordinal variables as well as nominal variables in the analysis.